RACE Regular Analysis Through Continuous Exercise

Alternating Currents (AC) : RACE-01

1. The current flowing through an ac circuit is given by

$I = 5\sin(120\pi t)\,A$

How long will the current take to reach the peak value starting from zero?

(A) $\frac{1}{60}\,s$ (B) 60s (C) $\frac{1}{120}\,s$ (D) $\frac{1}{240}\,s$

PAR143

2. The equation of current in a purely inductive circuit is $5\sin(49\pi t - 30°)$. If the inductance is 30 mH then the equation for the voltage across the inductor, will be :

$\left\{\text{Let } \pi = \frac{22}{7}\right\}$

(A) $1.47\sin(49\pi t - 30°)$ (B) $1.47\sin(49\pi t + 60°)$

(C) $23.1\sin(49\pi t - 30°)$ (D) $23.1\sin(49\pi t + 60°)$

PAR144

3. An AC source rated $220V$, 50 Hz is connected to a resistor. The time taken by the current to change from its maximum to the rms value is :

(A) 2.5 ms (B) 25 ms (C) 2.5 s (D) 0.25 ms

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PAR145

4. In LC circuit the inductance L = 40 mH and capacitance C = 100 μF. If a voltage $V(t)$ = 10 sin(314 t) is applied to the circuit, the current in the circuit is given as :

(A) 0.52 cos 314 t (B) 0.52 sin 314 t (C) 10 cos 314 t (D) 5.2 cos 314 t

PAR146

5. In an a. c. circuit, the instantaneous e.m.f. and current are given by

e = 100 sin 30 t

$i = 20\sin\left(30t - \frac{\pi}{4}\right)$

In one cycle of a.c., the average power consumed by the circuit and the wattless current are, respectively.

(A) $\frac{1000}{\sqrt{2}}$, 10 (B) $\frac{50}{\sqrt{2}}$, 0 (C) 50, 0 (D) 50, 10

PAR147

6. A circuit connected to an ac source of emf $e = e_0\sin(100t)$ with t in seconds, gives a phase difference of $\frac{\pi}{4}$ between the emf e and current i. Which of the following circuits will exhibit this?

(A) RC circuit with R = 1 $k\Omega$ and C = 1 μF (B) RL circuit with R = 1 $k\Omega$ and L = 1 mH

(C) RL circuit with R = 1 $k\Omega$ and L = 10 mH (D) RC circuit with R = 1 $k\Omega$ and C = 10 μF

PAR148

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7. An inductance coil has a reactance of 100 Ω. When an AC signal of frequency 1000 *Hz* is applied to the coil, the applied voltage leads the current by 45°. The self-inductance of the coil is :

(A) 1.1×10^{-2} *H* (B) 1.1×10^{-1} *H* (C) 5.5×10^{-5} *H* (D) 6.7×10^{-7} *H*

PAR149

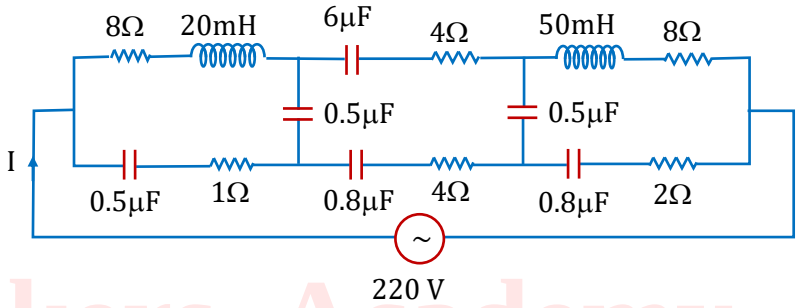
8. A 0.07 *H* inductor and a 12 Ω resistor are connected in series to a 220 *V*, 50 Hz ac source. The approximate current in the circuit and the phase angle between current and source voltage are respectively. [Take π as $\frac{22}{7}$]

(A) 8.8 *A* and $\tan^{-1}\left(\frac{11}{6}\right)$ (B) 88 *A* and $\tan^{-1}\left(\frac{11}{6}\right)$

(C) 0.88 *A* and $\tan^{-1}\left(\frac{11}{6}\right)$ (D) 8.8 *A* and $\tan^{-1}\left(\frac{6}{11}\right)$

PAR150

9. The effective current *I* in the given circuit at very high frequencies will be _____*A*



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(A) 22 (B) 33 (C) 44 (D) 55

PAR151

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	D	D	A	A	A	D	A	A	C

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RACE Regular Analysis Through Continuous Exercise

Alternating Currents (AC) : RACE-02

1. An AC circuit has $R = 100\ \Omega$, $C = 2\ \mu F$ and $L = 80\ mH$, connected in series. The quality factor of the circuit is :

(A) 0.5 (B) 2 (C) 20 (D) 400

PAR152

2. An LCR circuit contains resistance of 110 Ω and a supply of 220 V at 300 rad/s angular frequency. If only capacitance is removed from the circuit, current lags behind the voltage by 45°. If on the other hand, only inductor is removed the current leads by 45° with the applied voltage. The rms current flowing in the circuit will be :

(A) 1 A (B) 2.5 A (C) 1.5 A (D) 2A

PAR153

3. A series LCR circuit is designed to resonate at an angular frequency $\omega_0 = 10^5$ rad/s. The circuit draws 16 W power from 120 V source at resonance. The value of resistance 'R' in the circuit is ___Ω.

(A) 800 (B) 900 (C) 600 (D) 400

PAR154

4. Match **List-I** with **List-II** :

	List–I		**List–II**
(a)	$\omega L > \frac{1}{\omega C}$	(i)	Current is in phase with emf
(b)	$\omega L = \frac{1}{\omega C}$	(ii)	Current lags behind the applied emf
(c)	$\omega L < \frac{1}{\omega C}$	(iii)	Maximum current occurs
(d)	Resonant frequency	(iv)	Current leads the emf

Choose the **correct** answer from the options given below :

(A) (a) – (ii) ; (b) – (i) ; (c) - (iv) ; (d) – (iii) (B) (a) – (ii) ; (b) – (i) ; (c) - (iii) ; (d) – (iv)

(C) (a) – (iii) ; (b) – (i) ; (c) - (iv) ; (d) – (ii) (D) (a) – (iv) ; (b) – (iii) ; (c) - (ii) ; (d) – (i)

PAR155

5. In a series LCR circuit, the inductive reactance (X_L) is 10 Ω and the capacitive reactance (X_C) is 4 Ω. The resistance (R) in the circuit is 6 Ω. The power factor of the circuit is :

(A) $\frac{1}{2}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

PAR156

6. A block of mass 1 kg attached to a spring is made to oscillate with an initial amplitude of 12 cm. After 2 minutes the amplitude decreases to 6 cm. Determine the value of the damping constant for this motion. (take In2 = 0.693)

(A) $0.69 \times 10^2\ kg\ s^{-1}$ (B) $3.3 \times 10^2\ kg\ s^{-1}$ (C) $1.16 \times 10^{-2}\ kg\ s^{-1}$ (D) $5.7 \times 10^{-3}\ kg\ s^{-1}$

PAR157

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7. In a series LCR resonance circuit, if we change the resistance only, from a lower to higher value:

(A) The bandwidth of resonance circuit will increase.

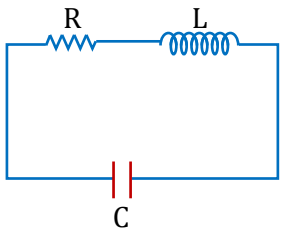
(B) The resonance frequency will increase.

(C) The quality factor will increase.

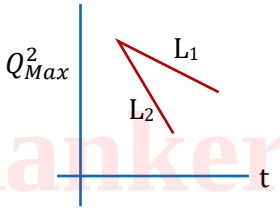
(D) The quality factor and the resonance frequency will remain constant.

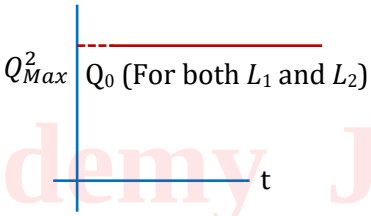
PAR158

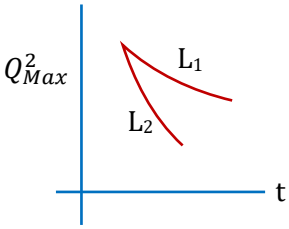
8. An LCR circuit is equivalent to a damped pendulum. In an LCR circuit the capacitor is charged to Q_0 and then connected to the L and R as shown below:

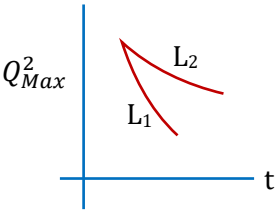


If a student plots graphs of the square of maximum charge (Q_{Max}^2) on the capacitor with time (t) for two different values L_1 and L_2 ($L_1 > L_2$) of L then which of the following represents this graph correctly? (Plots are schematic and not drawn to scale)

(A)


(B)


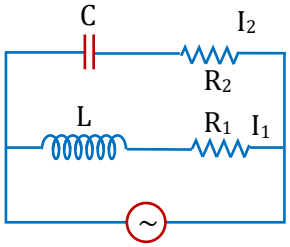
(C)


(D)


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PAR159

9. In the above circuit, $C = \frac{\sqrt{3}}{2}\mu F$, $R_2 = 20\Omega$, $L = \frac{\sqrt{3}}{10}H$ and $R_1 = 10\Omega$. Current in L-R_1 path is I_1 and in C-R_2 path it is I_2. The voltage of A.C source is given by $V = 200\sqrt{2}\sin(100t)$ volts. The phase difference between I_1 and I_2 is

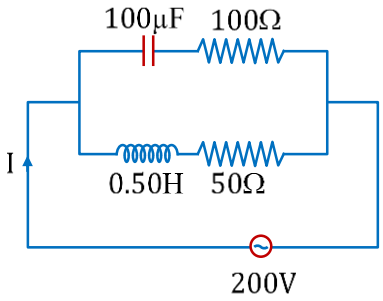


(A) 30° (B) 0° (C) 150° (D) 60°

PAR160

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10. In the given circuit the AC source has ω = 100 rad s^{-1}. Considering the inductor and capacitor to be ideal, what will be the current I flowing through the circuit?



(A) 5.9 A (B) 3.16 A (C) 0.94 A (D) 6 A

PAR161

11. A LCR circuit behaves like a damped harmonic oscillator. Comparing it with a physical spring-mass damped oscillator having damping constant 'b', the correct equivalence would be:

(A) $L \leftrightarrow m, C \leftrightarrow \frac{1}{k}, R \leftrightarrow b$ (B) $L \leftrightarrow \frac{1}{b}, C \leftrightarrow \frac{1}{m}, R \leftrightarrow \frac{1}{k}$

(C) $L \leftrightarrow m, C \leftrightarrow k, R \leftrightarrow b$ (D) $L \leftrightarrow k, C \leftrightarrow b, R \leftrightarrow m$

PAR162

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	B	D	B	A	C	C	A	C	C	B	A

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